

Basic Information

Experiment No.: 07

Date: 24-09-2026

Title: Java LinkedList and Its Methods

Objective / Aim

To implement "LinkedList" in Java and demonstrate insertion, accessing, removal, and queue-related operations.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

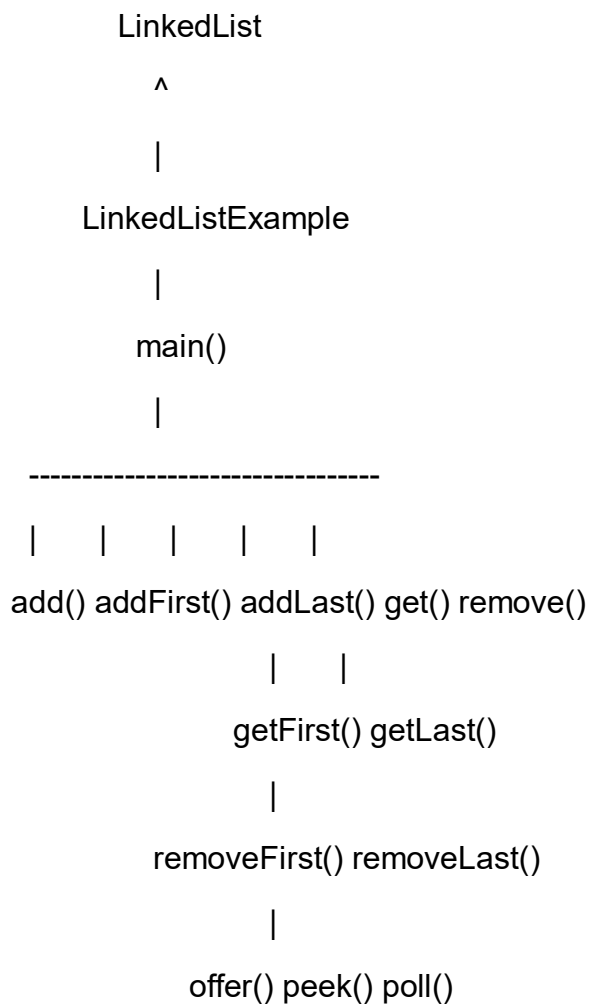
- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
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Algorithm / Logic Description

1. Create a "LinkedList" of strings.
2. Add elements using "add()".
3. Insert elements at the beginning and end using "addFirst()" and "addLast()".
4. Access elements using "get()", "getFirst()", and "getLast()".
5. Remove elements using "remove(index)" and "remove(Object)".

6. Remove the first and last elements using "removeFirst()" and "removeLast()".
 7. Add an element using "offer()".
 8. Access the first element using "peek()".
 9. Remove the first element using "poll()".
 10. Display the final list.
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UML Class Diagram



Source Code

```
package collectionframework;
```

```
import java.util.LinkedList;
```

```
public class LinkedListExample {
```

```
    public static void main(String[] args) {
```

```
        LinkedList<String> list = new LinkedList<>();
```

```
        // add()
```

```
        list.add("B");
```

```
        list.add("C");
```

```
        System.out.println("List: " + list);
```

```
        // addFirst()
```

```
        list.addFirst("A");
```

```
        // addLast()
```

```
        list.addLast("D");
```

```
        System.out.println("After addFirst/addLast: " + list);
```

```
        // get()
```

```
        System.out.println("Element at index 2: " + list.get(2));
```

```
        // getFirst()
```

```
        System.out.println("First: " + list.getFirst());
```

```
        // getLast()
```

```
System.out.println("Last: " + list.getLast());
```

```
// remove(index)
```

```
list.remove(1);
```

```
System.out.println("After remove index: " + list);
```

```
// remove(Object)
```

```
list.remove("C");
```

```
System.out.println("After remove object: " + list);
```

```
// Add elements again
```

```
list.addFirst("X");
```

```
list.addLast("Y");
```

```
// removeFirst()
```

```
System.out.println("Removed First: " + list.removeFirst());
```

```
// removeLast()
```

```
System.out.println("Removed Last: " + list.removeLast());
```

```
// offer()
```

```
list.offer("Mango");
```

```
System.out.println("After offer: " + list);
```

```
// peek()
```

```
System.out.println("Peek: " + list.peek());
```

```
// poll()
```

```
System.out.println("Poll: " + list.poll());
```

```
        System.out.println("Final List: " + list);
    }
}
```

Compilation & Execution Commands

Save the file as:

LinkedListExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

javac LinkedListExample.java

Run:

java collectionframework.LinkedListExample

Output Screen

List: [B, C]

After addFirst/addLast: [A, B, C, D]

Element at index 2: C

First: A

Last: D

After remove index: [A, C, D]

After remove object: [A, D]

Removed First: X

Removed Last: Y

After offer: [A, D, Mango]

Peek: A

Poll: A

Final List: [D, Mango]

Result

The "LinkedList" class was successfully implemented, and insertion, accessing, removal, and queue-related operations such as "offer()", "peek()", and "poll()" were successfully demonstrated.